AGGREGATE Function in Excel

1. What is AGGREGATE?

The AGGREGATE function calculates a summary of a list, such as a sum, average, count, maximum, median, or k-th largest value, and lets you choose what to ignore: hidden rows, error values, or both. It solves the problem that SUM and AVERAGE fail when one cell has an error, and SUBTOTAL cannot skip errors. A Data Analyst uses it to get clean results from messy data without helper columns.

2. Syntax

Reference form: `=AGGREGATE(function_num, options, ref1, [ref2], ...)`
Array form: `=AGGREGATE(function_num, options, array, [k])`

Argument	Required?	What it accepts
`function_num`	Required	A number from 1 to 19 that chooses the calculation (see the first table below)
`options`	Required	A number from 0 to 7 that chooses what to ignore (see the second table below)
`ref1 / array`	Required	A range, cell reference, or array to calculate (for example `B2:B6`)
`ref2, ... / k`	Optional	Reference form: more ranges, up to 253. Array form: k, which functions 14 to 19 need.

#	Function	#	Function	#	Function	#	Function	#	Function
1	AVERAGE	5	MIN	9	SUM	13	MODE.SNGL	17	QUARTILE.INC
2	COUNT	6	PRODUCT	10	VAR.S	14	LARGE	18	PERCENTILE.EXC
3	COUNTA	7	STDEV.S	11	VAR.P	15	SMALL	19	QUARTILE.EXC
4	MAX	8	STDEV.P	12	MEDIAN	16	PERCENTILE.INC		

Option	0	1	2	3	4	5	6	7
Ignores hidden rows	No	Yes	No	Yes	No	Yes	No	Yes
Ignores error values	No	No	Yes	Yes	No	No	Yes	Yes
Ignores nested SUBTOTAL / AGGREGATE	Yes	Yes	Yes	Yes	No	No	No	No

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	70
3	Karim	80
4	Hasan	#N/A
5	Nabil	90
6	**Total**	**240**

Formula: `=AGGREGATE(9, 6, B2:B5)`
Function 9 is SUM and option 6 ignores errors. Excel skips #N/A and adds 70 + 80 + 90. **Result: 240**. A plain SUM would return #N/A.

4. Another Example

	A	B
1	Product	Monthly Sales
3	Mouse	3500
4	Keyboard	4500
5	Monitor	8000
6	Headset	6000
7	**2nd Largest**	**6000**

Formula: `=AGGREGATE(14, 5, B2:B6, 2)`
Function 14 is LARGE, option 5 ignores hidden rows, and k = 2. A filter hides row 2 (Laptop, 12000), so the 2nd largest visible value is 6000. **Result: 6000**. A plain LARGE would give 8000.

5. When to Use It

- Sum or average a column that contains errors such as #N/A or #DIV/0!
- Find the largest, smallest, or k-th value among the visible rows only
- Get the median, percentile, or quartile of filtered data
- Count the numbers in a range while skipping errors and hidden rows

6. Important Things to Remember

- Functions 1 to 13 use the reference form. Functions 14 to 19 (LARGE, SMALL, PERCENTILE, QUARTILE) need k.
- Hidden rows means filtered rows and rows hidden by hand. Hidden columns are never ignored.
- Options 0 to 3 ignore nested SUBTOTAL and AGGREGATE formulas. Options 4 to 7 count them. Empty options works like 0.
- If the option you choose does not ignore an error, the result shows that error. An invalid function_num or option, a missing k, or a 3-D reference returns `#VALUE!`.

7. Common Mistakes

Mistake: Leaving options empty or typing 0 and expecting errors to be skipped. **Fix:** Use 6 to ignore errors, or 7 for hidden rows and errors.

Mistake: Using function 14 to 19 without k, such as `=AGGREGATE(14, 6, B2:B6)`. **Fix:** Add k: `=AGGREGATE(14, 6, B2:B6, 2)`.

Mistake: Expecting hidden columns to be ignored. No option skips them. **Fix:** Filter rows instead.

Mistake: Skipping errors without asking why they exist. A skipped error can hide bad data. **Fix:** Resolve the source when you can, and count what was skipped (see the tip).

8. Data Analyst Tip

Always compare the result with the data it came from. Next to it, calculate `=AGGREGATE(2, 7, B2:B6)` to see how many numbers were used, and compare it with `=ROWS(B2:B6)`. If it is lower, some cells are errors, text, or hidden, so the result covers part of the data.